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Aircraft powerplant gearbox with selectable power coupler

Abstract

An aircraft powerplant assembly includes a gearbox and a power coupler. The gearbox includes a first power transfer apparatus and a second power transfer apparatus independent of the first power transfer apparatus. The power coupler includes a clutch. The first power transfer apparatus is configured to operatively couple and transfer mechanical power through the gearbox between a first engine apparatus and an engine accessory. The power coupler is configured to operatively couple a power coupled accessory to the second power transfer apparatus when the clutch is engaged. The second power transfer apparatus and the power coupler are collectively configured to operatively couple and transfer mechanical power through the gearbox between a second engine apparatus and the power coupled accessory when the clutch is engaged. The power coupler is configured to operatively decouple the second power transfer apparatus from the power coupled accessory when the clutch is disengaged.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

1. Technical Field

(1) This disclosure relates generally to an aircraft powerplant and, more particularly, to an accessory system for the aircraft powerplant.

2. Background Information

(2) An aircraft powerplant such as a turbofan engine includes an accessory system. This accessory system may include various accessories such as fluid pumps, electric motors, electric generators, and the like coupled to an accessory gearbox. Various types and configurations of accessory systems are known in the art. While these known accessory systems have various benefits, there is still room in the art for improvement.

SUMMARY OF THE DISCLOSURE

(3) According to an aspect of the present disclosure, an assembly is provided for an aircraft powerplant. This assembly includes a gearbox and a power coupler. The gearbox includes a first

power transfer apparatus and a second power transfer apparatus independent of the first power transfer apparatus. The power coupler is mounted to an exterior of the gearbox and includes a clutch. The first power transfer apparatus is configured to operatively couple and transfer mechanical power through the gearbox between a first engine apparatus and an engine accessory. The power coupler is configured to operatively couple a power coupled accessory to the second power transfer apparatus when the clutch is engaged. The second power transfer apparatus and the power coupler are collectively configured to operatively couple and transfer mechanical power through the gearbox between a second engine apparatus and the power coupled accessory when the clutch is engaged. The power coupler is configured to operatively decouple the second power transfer apparatus from the power coupled accessory when the clutch is disengaged.

(4) According to another aspect of the present disclosure, another assembly is provided for an aircraft powerplant. This assembly includes a first rotating structure, a second rotating structure, an engine accessory, a power coupled accessory, a gearbox and a power coupler. The first rotating structure includes a first bladed engine rotor. The second rotating structure includes a second bladed engine rotor. The gearbox includes a first power transfer apparatus and a second power transfer apparatus that is mechanically disengaged from the first power transfer apparatus. The power coupler is between the power coupled accessory and the gearbox. The first power transfer apparatus is configured to operatively couple and transfer mechanical power through the gearbox between the first rotating structure and the engine accessory. During a first operating mode, the power coupler is configured to operatively couple the power coupled accessory to the second power transfer apparatus, and the second power transfer apparatus and the power coupler are collectively configured to operatively couple and transfer mechanical power through the gearbox between the second rotating structure and the power coupled accessory. During a second operating mode, the power coupler is configured to operatively decouple the second power transfer apparatus from the power coupled accessory.

(5) According to still another aspect of the present disclosure, another assembly is provided for an aircraft powerplant. This assembly includes a first rotating structure, a second rotating structure, an engine accessory, a power coupled accessory, a gearbox and a power coupler. The first rotating structure includes a first bladed engine rotor. The second rotating structure includes a second bladed engine rotor. The gearbox includes a first power transfer apparatus and a second power transfer apparatus that is mechanically disengaged from the first power transfer apparatus. The power coupler is arranged with the gearbox. The gearbox is disposed between the power coupled accessory and the power coupler. The first power transfer apparatus is configured to operatively couple and transfer mechanical power through the gearbox between the first rotating structure and the engine accessory. During a first operating mode, the power coupler is configured to operatively couple the power coupled accessory to the second power transfer apparatus, and the second power transfer apparatus and the power coupler are collectively configured to operatively couple and transfer mechanical power through the gearbox between the second rotating structure and the power coupled accessory. During a second operating mode, the power coupler is configured to operatively decouple the second power transfer apparatus from the power coupled accessory.

(6) The power coupler may include a clutch configured to: engage during the first operating mode; and disengage during the second operating mode.

(7) The power coupler may be mounted to an exterior of the gearbox. The power coupled accessory may be mounted to an exterior of the power coupler.

(8) The power coupler may be integrated with the power coupled accessory into a module that is removably coupled to an exterior of the gearbox.

(9) The power coupler may be configured to engage the clutch when a rotational speed of the second engine apparatus is equal to or less than a threshold. The power coupler may be configured to disengage the clutch when the rotational speed of the second engine apparatus is greater than the threshold.

(10) The power coupler may also include a gear system configured to transfer mechanical power between the power coupled accessory and the second power transfer apparatus when the clutch is engaged.
(11) The assembly may also include the power coupled accessory mounted to the gearbox through the power coupler.
(12) The assembly may also include the power coupled accessory mounted to the exterior of the gearbox independent of the power coupler.
(13) The power coupler may be mounted to a first side of the gearbox. The power coupled accessory may be mounted to a second side of the gearbox opposite the first side of the gearbox.
(14) The gearbox may also include a third power transfer apparatus independent of the first power transfer apparatus and the second power transfer apparatus. The third power transfer apparatus may be configured to operatively couple and transfer mechanical power through the gearbox between the power coupler and the power coupled accessory.
(15) The third power transfer apparatus may be configured as a pass-through shaft extending through the gearbox.
(16) The assembly may also include the power coupled accessory and a housing mounted to the exterior of the gearbox. The power coupler and the power coupled accessory may be housed within the housing.
(17) The assembly may also include the power coupled accessory. The power coupler and the power coupled accessory may be configured together in a single line replaceable unit that is removably mounted to the gearbox.
(18) The assembly may also include the power coupled accessory. The power coupled accessory may include an electric machine.
(19) The assembly may also include the power coupled accessory. The power coupled accessory may be configurable as at least one of an electric motor or an electric generator.
(20) The second power transfer apparatus may be configured to operatively couple and transfer mechanical power through the gearbox between the second engine apparatus and a third engine accessory independent of the power coupler.
(21) The gearbox may also include a gearbox housing. The second power transfer apparatus may be configured as or otherwise include a gear system disposed in the gearbox housing.
(22) The gearbox may also include a gearbox housing. The first power transfer apparatus may be configured as or otherwise include a first gear system disposed in the gearbox housing. The second power transfer apparatus may be configured as or otherwise include a second gear system disposed in the gearbox housing.
(23) The assembly may also include a compressor section, a combustor section, a first turbine section, a second turbine section and a flowpath extending through the compressor section, the combustor section, the first turbine section and the second turbine section from an inlet into the flowpath to an exhaust from the flowpath. The first engine apparatus may include a first turbine rotor disposed in the first turbine section. The second engine apparatus may include a second turbine rotor disposed in the second turbine section.
(24) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.
(25) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a partial schematic sectional illustration of an aircraft propulsion system.

(2) FIG. **2** is a schematic illustration of a portion of the aircraft propulsion system at an accessory system.
(3) FIGS. **3** and **4** are schematic illustrations of a portion of the aircraft propulsion system at the accessory system with various alternative power coupler arrangements.
(4) FIG. **5** is a schematic illustration of a portion of the aircraft propulsion system with an electric machine electrically coupled to an electrical system.

DETAILED DESCRIPTION

(5) FIG. **1** illustrates a powerplant **20** for an aircraft. The aircraft may be an airplane, a rotorcraft (e.g., a helicopter), a drone (e.g., an unmanned aerial vehicle (UAV)) or any other manned or unmanned aerial vehicle or system. For ease of description, the aircraft powerplant **20** is described below as a propulsion system **22** for the aircraft and, more particularly, as a turbofan propulsion system. The aircraft powerplant **20** of the present disclosure, however, is not limited to such an exemplary propulsion system. The aircraft propulsion system **22**, for example, may alternatively be configured as a turbojet propulsion system, a turboprop propulsion system, a turboshaft propulsion system, a propfan propulsion system, a pusher fan propulsion system, or any other type of ducted and/or open rotor propulsion system. Moreover, the aircraft powerplant **20** is not limited to propulsion system applications. The aircraft powerplant **20**, for example, may alternatively (or also) be configured as an electrical power system for the aircraft; e.g., an auxiliary power unit (APU).
(6) The aircraft propulsion system **22** includes a gas turbine engine **24** (e.g., a turbofan engine) housed within a stationary engine housing **26**, which engine housing **26** of FIG. **1** includes an inner housing structure **28**, an outer housing structure **30** and a vane structure **31** (e.g., a fan exit guide vane (FEGV) structure) extending radially between and connected to the inner housing structure **28** and the outer housing structure **30**. The aircraft propulsion system **22** also includes an accessory system **32** servicing one or more components and/or sub-systems of the aircraft propulsion system **22** and its turbine engine **24**. The aircraft propulsion system **22** extends axially along an axis **34** between an axial forward, upstream end **36** of the aircraft propulsion system **22** and an axial aft, downstream end **38** of the aircraft propulsion system **22**. Briefly, the powerplant axis **34** may be a centerline axis of the aircraft propulsion system **22**, the turbine engine **24** and/or one or more of its members. The powerplant axis **34** may also or alternatively be a rotational axis for one or more members of the turbine engine **24**.
(7) The aircraft propulsion system **22** and its turbine engine **24** of FIG. **1** includes a propulsor section **40** (e.g., a fan section), a compressor section **41**, a combustor section **42** and a turbine section **43**. The compressor section **41** of FIG. **1** includes a low pressure compressor (LPC) section **41**A and a high pressure compressor (HPC) section **41**B. The turbine section **43** of FIG. **1** includes a high pressure turbine (HPT) section **43**A and a low pressure turbine (LPT) section **43**B. Here, at least (or only) the LPC section **41**A, the HPC section **41**B, the combustor section **42**, the HPT section **43**A and the LPT section **43**B collectively form a core **46** of the turbine engine **24**.
(8) The engine sections **40-43**B may be arranged sequentially along the powerplant axis **34** within the engine housing **26**. The propulsor section **40** includes a bladed propulsor rotor **48**; e.g., a fan rotor. The LPC section **41**A includes a bladed low pressure compressor (LPC) rotor **49**. The HPC section **41**B includes a bladed high pressure compressor (HPC) rotor **50**. The HPT section **43**A includes a bladed high pressure turbine (HPT) rotor **51**. The LPT section **43**B includes a bladed low pressure turbine (LPT) rotor **52**. The propulsor rotor **48**, the LPC rotor **49**, the HPC rotor **50**, the HPT rotor **51** and the LPT rotor **52** each include a rotor base (e.g., a disk or a hub) and a plurality of rotor blades (e.g., airfoils, rotor vanes, etc.). The rotor blades are arranged and may be equispaced circumferentially around the respective rotor base in one or more arrays. With this arrangement, the rotor blades may be arranged into one or more stages. Each of the rotor blades is connected to (e.g., formed integral with or otherwise attached to) the respective rotor base. Each of the rotor blades projects radially (e.g., spanwise) out from the respective rotor base to a distal tip of

the respective rotor blade.

(9) The HPC rotor **50** is coupled to and rotatable with the HPT rotor **51**. The HPC rotor **50** of FIG. **1**, for example, is connected to the HPT rotor **51** through a high speed shaft **54**. At least (or only) the HPC rotor **50**, the HPT rotor **51** and the high speed shaft **54** collectively form a high speed rotating structure **56**; e.g., a high speed spool of the engine core **46**. This high speed rotating structure **56** of FIG. **1** and its members **50**, **51** and **54** are rotatable about the powerplant axis **34**. However, it is contemplated the high speed rotating structure **56** may alternatively be rotatable about another axis radially and/or angularly offset from the rotational axis of the propulsor rotor **48** and/or the centerline axis of the turbine engine **24**.

(10) The LPC rotor **49** is coupled to and rotatable with the LPT rotor **52**. The LPC rotor **49** of FIG. **1**, for example, is connected to the LPT rotor **52** through a low speed shaft **58**. At least (or only) the LPC rotor **49**, the LPT rotor **52** and the low speed shaft **58** collectively form a low speed rotating structure **60**; e.g., a low speed spool of the engine core **46**. This low speed rotating structure **60** is further coupled to the propulsor rotor **48** through a drivetrain **62**. The drivetrain **62** may be configured as a geared drivetrain, where a geartrain **64** (e.g., a transmission, a speed change device, an epicyclic geartrain, etc.) is disposed between and operatively couples the propulsor rotor **48** to the low speed rotating structure **60** and its LPT rotor **52**. With this arrangement, the propulsor rotor **48** may rotate at a different (e.g., slower) rotational speed than the low speed rotating structure **60** and its LPT rotor **52**. Alternatively, the drivetrain **62** may be configured as a direct-drive drivetrain, where the geartrain **64** is omitted. With such an arrangement, the propulsor rotor **48** rotates at a common (the same) rotational speed as the low speed rotating structure **60** and its LPT rotor **52**. The low speed rotating structure **60** of FIG. **1** and its members **49**, **52** and **58** as well as the propulsor rotor **48** are rotatable about the powerplant axis **34**. However, it is contemplated the low speed rotating structure **60** may alternatively be rotatable about another axis radially and/or angularly offset from the rotational axis of the propulsor rotor **48** and/or the centerline axis of the turbine engine **24**.

(11) The inner housing structure **28** of FIG. **1** includes an inner case **66** (e.g., a core case) for the turbine engine **24**, an inner nacelle structure **68** (sometimes referred to as an inner fixed structure (IFS)) and an internal inner housing compartment **70**. The inner case **66** is disposed radially outboard of, extends axially along and may circumscribe one or more or all of the engine sections **41**A-**43**B and their respective bladed engine rotors **49-52**. The inner case **66** may thereby house and provide a support structure for the respective bladed engine sections **41**A-**43**B and their respective engine rotors **49-52**. The inner nacelle structure **68** is configured to provide an aerodynamic cover over the engine core **46** and its inner case **66**. The inner housing compartment **70** of FIG. **1** is formed by and is disposed radially between the inner case **66** and an inner barrel of the inner nacelle structure **68**. The inner housing structure **28** and its inner nacelle structure **68** may also form a radial inner peripheral boundary of a bypass flowpath **72** (e.g., an annular bypass flowpath) within the aircraft propulsion system **22**.

(12) The outer housing structure **30** of FIG. **1** includes an outer case **74** (e.g., a fan case) for the turbine engine **24** and an outer nacelle structure **76**. The outer case **74** is disposed radially outboard of, extends axially along and may circumscribe the propulsor section **40** and its propulsor rotor **48**. The outer case **74** may thereby house and provide a containment structure for the propulsor section **40** and its propulsor rotor **48**. The outer nacelle structure **76** is configured to provide an aerodynamic cover over the outer case **74**. The outer housing structure **30** and its outer nacelle structure **76** may also form a radial outer peripheral boundary of the bypass flowpath **72**.

(13) During operation, ambient air from outside of the aircraft enters the aircraft propulsion system **22** and its turbine engine **24** through an airflow inlet **78**. This air is directed across the propulsor section **40** and into a core flowpath **80** (e.g., annular core flowpath) and the bypass flowpath **72**. The core flowpath **80** of FIG. **1** extends sequentially through the LPC section **41**A, the HPC section **41**B, the combustor section **42**, the HPT section **43**A and the LPT section **43**B from an airflow inlet

82 into the core flowpath **80** to a combustion products exhaust **84** out from the core flowpath **80** and the engine core **46**. The air entering the core flowpath **80** may be referred to as "core air". The bypass flowpath **72** extends through a bypass duct, which bypass flowpath **72** and bypass duct bypass (e.g., are disposed radially outboard of and extend along) the engine core **46** and the inner housing structure **28**. The air within the bypass flowpath **72** may be referred to as "bypass air".
(14) The core air is compressed by the LPC rotor **49** and the HPC rotor **50** and is directed into a combustion chamber **86** (e.g., an annular combustion chamber) of a combustor (e.g., an annular combustor) in the combustor section **42**. Fuel is injected into the combustion chamber **86** by one or more fuel injectors and mixed with the compressed core air to provide a fuel-air mixture. This fuel-air mixture is ignited and combustion products thereof flow through and sequentially drive rotation of the HPT rotor **51** and the LPT rotor **52** about the powerplant axis **34**. The rotation of the HPT rotor **51** and the LPT rotor **52** respectively drive rotation of the HPC rotor **50** and the LPC rotor **49** about the powerplant axis **34** and, thus, compression of the air received from the core inlet **82**. The rotation of the LPT rotor **52** also drives rotation of the propulsor rotor **48**. The rotation of the propulsor rotor **48** propels the bypass air through and out of the bypass flowpath **72**. The propulsion of the bypass air may account for a majority of thrust generated by the turbine engine **24** of FIG. **1**. Briefly, within the bypass flowpath **72**, the vane structure **31** conditions (e.g., straightens out, de-swirls, etc.) the flow of bypass air propelled by the propulsor rotor **48** to enhance the forward thrust.
(15) While the turbine engine **24** is described above with a particular two rotating structure arrangement, the present disclosure is not limited thereto. For example, the LPC rotor **49** may be omitted to configure the LPT rotor **52** as a power turbine (PT) rotor for the propulsor rotor **48**. In another example, the turbine engine **24** may also include another rotating structure; e.g., an intermediate speed spool for the engine core **46**.
(16) Referring to FIG. **2**, the accessory system **32** includes a set of one or more gearbox mounted engine accessories, an accessory gearbox **88** and a power coupler **90** for the accessory gearbox **88**. The gearbox mounted engine accessories of FIG. **2** includes one or more first engine accessories **92**A-E (generally referred to as "**92**"), a second engine accessory **94** and a third engine accessory **96**. The present disclosure, however, is not limited to such an exemplary gearbox mounted engine accessory arrangement. For example, the accessory system **32** may be configured with multiple second engine accessories **94** and/or multiple third engine accessories **96**. Moreover, it is contemplated the accessory system **32** may alternatively be configured without the first engine accessories **92** and/or the second engine accessory **94**.
(17) Each of the first engine accessories **92**A-E includes a first engine accessory housing **98**A-E (generally referred to as "**98**") (e.g., a case) and a first engine accessory rotor **100**A-E (generally referred to as "**100**") at least partially housed within the respective first engine accessory housing **98**. Each first engine accessory **92** is mounted to or otherwise arranged with the accessory gearbox **88**, for example external to the accessory gearbox **88**. The first engine accessory housing **98** of each first engine accessory **92**, for example, may be mechanically fastened and/or otherwise attached to an exterior of a housing **102** (e.g., a case) of the accessory gearbox **88**. Briefly, the gearbox housing **102** of FIG. **2** extends axially along the powerplant axis **34** (see FIG. **1**) between opposing axial sides **104** and **106** of the accessory gearbox **88** and its gearbox housing **102**. The gearbox housing **102** of FIG. **2** also extends laterally (e.g., circumferentially about the powerplant axis **34** and/or tangentially to a reference circle extending around the powerplant axis **34**) between opposing lateral ends **108** and **110** of the accessory gearbox **88** and its gearbox housing **102**. The first engine accessories **92**A-C may be disposed at (e.g., on, adjacent or proximate) the gearbox first side **104**. The first engine accessories **92**A-C of FIG. **2**, for example, are arranged laterally along the gearbox first side **104** between the gearbox first end **108** and the third engine accessory **96**. The first engine accessories **92**D and **92**E may be disposed at the gearbox second side **106**. The first engine accessories **92**D and **92**E of FIG. **2**, for example, are arranged laterally along the gearbox first side

104 between the gearbox first end **108** and the power coupler **90**. The present disclosure, however, is not limited to such an exemplary arrangement. All of the first engine accessories **92**, for example, may be disposed to the gearbox first side **104** or the gearbox second side **106**.
(18) The second engine accessory **94** includes a second engine accessory housing **112** (e.g., a case) and a second engine accessory rotor **114** at least partially housed within the second engine accessory housing **112**. The second engine accessory **94** is mounted to or otherwise arranged with the accessory gearbox **88**, for example external to the accessory gearbox **88**. The second engine accessory housing **112**, for example, may be mechanically fastened and/or otherwise attached to the gearbox housing **102**. The second engine accessory **94** may be disposed at the gearbox first side **104**. The second engine accessory **94** of FIG. **2**, for example, is arranged laterally along the gearbox first side **104** between the gearbox second end **110** and the third engine accessory **96**. The present disclosure, however, is not limited to such an exemplary arrangement. For example, the second engine accessory **94** may alternatively be arranged laterally along the gearbox second side **106** between the gearbox second end **110** and the power coupler **90**.
(19) The third engine accessory **96** includes a third engine accessory housing **116** (e.g., a case) and a third engine accessory rotor **118** at least partially housed within the third engine accessory housing **116**. The third engine accessory **96** is mounted to or otherwise arranged with the accessory gearbox **88**, for example external to the accessory gearbox **88**. The third engine accessory housing **116**, for example, may be mechanically fastened and/or otherwise attached to the gearbox housing **102**. The third engine accessory **96** may be disposed at the gearbox first side **104**. The third engine accessory **96** of FIG. **2**, for example, is arranged laterally along the gearbox first side **104** between the first engine accessories **98**A-C and the second engine accessory **94**. Here, the third engine accessory **96** is laterally aligned with and axially opposite the power coupler **90**.
(20) Examples of the engine accessories **92**, **94**, **96** include, but are not limited to, fluid pump(s), fluid conditioner(s) and/or electric machine(s). Examples of the fluid pumps include, but are not limited to, fuel pump(s), hydraulic pump(s) and/or lubricant pump(s). Examples of the fluid conditioners include, but are not limited to, de-oiler(s) and/or separator(s). Examples of the electric machines include, but are not limited to, variable frequency generator(s), integral drive generator(s), permanent magnet motor-generator(s) and/or permanent magnet motor(s).
(21) The accessory gearbox **88** includes the gearbox housing **102**, a first engine power transfer apparatus **120**, a second engine power transfer apparatus **122** and an accessory power transfer apparatus **124**. Each of the power transfer apparatuses **120**, **122**, **124** is partially or completely housed within the accessory gearbox **88** and its gearbox housing **102**. The power transfer apparatuses **120**, **122** and **124** of FIG. **2**, for example, may be partially or completely located within a common (e.g., the same) internal gearbox compartment **126** of the gearbox housing **102**. Alternatively, the gearbox compartment **126** may be sub-divided into one or more (e.g., fluidly coupled) sub-compartments within the gearbox housing **102** by one or more baffles. Still alternatively, the gearbox housing **102** may include multiple internal gearbox compartments, where at least one of the power transfer systems **120**, **122**, **124** may be partially or completely located within a separate one of the gearbox compartments than another one of the power transfer systems **120**, **122**, **124**.
(22) The first engine power transfer apparatus **120** is operatively independent of the other power transfer apparatuses **122** and **124** within the accessory gearbox **88** and its gearbox housing **102**. The first engine power transfer apparatus **120** may also be operatively independent of the other power transfer apparatuses **122** and **124** outside the accessory gearbox **88** and its gearbox housing **102**. The first engine power transfer apparatus **120** of FIG. **2**, for example, is spatially separated from and mechanically decoupled from the other power transfer apparatuses **122** and **124** both within and outside of the accessory gearbox **88** and its gearbox housing **102**.
(23) The first engine power transfer apparatus **120** of FIG. **2** includes a first engine coupling **128**, one or more first engine accessory couplings **130**A-E (generally referred to as "**130**") and a first

gearbox gear system **132**. The first engine coupling **128** is operatively coupled to a first engine apparatus **134** (e.g., an engine rotating structure, an engine rotor, etc.) through, for example, a first drivetrain **136**. This first drivetrain **136** may be configured as or otherwise include a shaft, a tower shaft assembly, another gearbox (e.g., an angle gearbox), and/or the like. For case of description, the first engine apparatus **134** may be described below as the high speed rotating structure **56**. However, in other embodiments, it is contemplated the first engine apparatus **134** may alternatively be the low speed rotating structure **60** or another rotating structure or rotor within the turbine engine **24**. Each first engine accessory coupling **130** of FIG. **2** is operatively coupled to the first engine accessory rotor **100** of a respective one of the first engine accessories **92**. The first gearbox gear system **132** is (e.g., completely) housed within the accessory gearbox **88**. This first gearbox gear system **132** is operatively coupled between and operatively interconnects the various first engine power transfer apparatus couplings **128** and **130**.

(24) The second engine power transfer apparatus **122** is operatively independent of the other power transfer apparatuses **120** and **124** within the accessory gearbox **88** and its gearbox housing **102**. The second engine power transfer apparatus **122** may also be operatively independent of the first engine power transfer apparatus **120** outside the accessory gearbox **88** and its gearbox housing **102**. The second engine power transfer apparatus **122** of FIG. **2**, for example, is spatially separated from and mechanically decoupled from the other power transfer apparatuses **120** and **124** within of the accessory gearbox **88** and its gearbox housing **102**. The second engine power transfer apparatus **122** of FIG. **2** is also spatially separated from and mechanically decoupled from the first engine power transfer apparatus **120** outside of the accessory gearbox **88** and its gearbox housing **102**.

(25) The second engine power transfer apparatus **122** of FIG. **2** includes a second engine coupling **138**, a second engine accessory coupling **140**, a first power coupler coupling **142** and a second gearbox gear system **144**. The second engine coupling **138** is operatively coupled to a second engine apparatus **146** (e.g., an engine rotating structure, an engine rotor, etc.) through, for example, a second drivetrain **148**. This second drivetrain **148** may be configured as or otherwise include a shaft, a tower shaft assembly, another gearbox (e.g., an angle gearbox), and/or the like. For case of description, the second engine apparatus **146** may be described below as the low speed rotating structure **60**. However, in other embodiments, it is contemplated the second engine apparatus **146** may alternatively be the high speed rotating structure **56** or another rotating structure or rotor within the turbine engine **24**. The second engine accessory coupling **140** of FIG. **2** is operatively coupled to the second engine accessory rotor **114**. The second gearbox gear system **144** is (e.g., completely) housed within the accessory gearbox **88**. This second gearbox gear system **144** is operatively coupled between and operatively interconnects the various second engine power transfer apparatus couplings **138**, **140** and **142**.

(26) The accessory power transfer apparatus **124** is operatively independent of the engine power transfer apparatuses **120** and **122** within the accessory gearbox **88** and its gearbox housing **102**. The accessory power transfer apparatus **124** of FIG. **2**, for example, is spatially separated from and mechanically decoupled from the engine power transfer apparatuses **120** and **122** within the accessory gearbox **88** and its gearbox housing **102**. The accessory power transfer apparatus **124** may also be operatively independent of the first engine power transfer apparatus **120** outside the accessory gearbox **88** and its gearbox housing **102**. By contrast, the accessory power transfer apparatus **124** of FIG. **2** is selectively operatively coupled to the second engine power transfer apparatus **122** outside of the accessory gearbox **88** through the power coupler **90** as described below.

(27) The accessory power transfer apparatus **124** of FIG. **2** includes a third engine accessory coupling **150**, a second power coupler coupling **152** and a pass-through shaft **154**. The third engine accessory coupling **150** of FIG. **2** is operatively coupled to the third engine accessory rotor **118**. The pass-through shaft **154** is operatively coupled between and operatively interconnects the accessory power transfer apparatus couplings **150** and **152**. The third engine accessory coupling

150 of FIG. **2**, for example, is connected to (or integrated as part of) the pass-through shaft **154** at a first end of the pass-through shaft **154**. The second power coupler coupling **152** of FIG. **2** is connected to (or integrated as part of) the pass-through shaft **154** at a second end of the pass-through shaft **154**, where the second end of the pass-through shaft **154** is opposite the first end of the pass-through shaft **154**. Here, the pass-through shaft **154** passes axially through the accessory gearbox **88** and its gearbox housing **102**. This pass-through shaft **154** is also arranged laterally between the first gearbox gear system **132** and the second gearbox gear system **144** within the accessory gearbox **88** and its gearbox housing **102**.
(28) The power coupler **90** of FIG. **2** includes a coupler housing **156** (e.g., a case), a first coupler assembly **158** and a second coupler assembly **160**, where each coupler assembly **158**, **160** is at least partially or completely housed within the coupler housing **156**. The power coupler **90** is removably mounted to or otherwise arranged with the accessory gearbox **88**. The coupler housing **156**, for example, may be mechanically fastened and/or otherwise attached to the gearbox housing **102**. The power coupler **90** may be disposed at the gearbox second side **106**. The power coupler **90** of FIG. **2**, for example, is arranged laterally along the gearbox second side **106** between the fifth engine accessory **92**E and the gearbox second end **110**. Here, the power coupler **90** is laterally aligned with and axially opposite the third engine accessory **96**. The power coupler **90** may also laterally overlap one or more of the power transfer apparatuses **120** and/or **122**.
(29) The first coupler assembly **158** includes a first coupler shaft **162**, a first coupler clutch **164** and a first coupler gear **166**. The first coupler shaft **162** is operatively coupled to the second engine power transfer apparatus **122**. More particularly, the first coupler shaft **162** of FIG. **2** is connected to and rotatable with the first power coupler coupling **142**. The first coupler clutch **164** is between the first coupler shaft **162** and the first coupler gear **166**. This first coupler clutch **164** is configured to selectively (a) mechanically connect the first coupler shaft **162** to the first coupler gear **166** such that the first coupler shaft **162** is rotatable with the first coupler gear **166**, or (b) mechanically disconnect the first coupler shaft **162** from the first coupler gear **166** such that the first coupler shaft **162** is rotatably independent from the first coupler gear **166** and vice versa.
(30) The second coupler assembly **160** includes a second coupler shaft **168** and a second coupler gear **170**. The second coupler shaft **168** is operatively coupled to the accessory power transfer apparatus **124**. More particularly, the second coupler shaft **168** of FIG. **2** is connected to and rotatable with the second power coupler coupling **152**. The second coupler gear **170** is connected to (e.g., mounted on) and rotatable with the second coupler shaft **168**. The second coupler gear **170** is meshed with the first coupler gear **166**.
(31) During aircraft powerplant operation, the power coupler **90** is configured to facilitate a selective operational coupling of the second engine apparatus **146** and the third engine accessory rotor **118**. For example, during a first mode of operation, the power coupler **90** of FIG. **2** may operatively couple and transfer mechanical power between the accessory power transfer apparatus **124** and the second engine power transfer apparatus **122**. Here, the power coupler **90** and the gearbox members **122** and **124** are collectively configured to operatively couple and transfer the mechanical power between the second engine apparatus **146** and the third engine accessory rotor **118**. During a second mode of operation, the power coupler **90** of FIG. **2** may operatively decouple the accessory power transfer apparatus **124** from the second engine power transfer apparatus **122**. Here, the second engine apparatus **146** is operatively decoupled from the third engine accessory rotor **118**.
(32) During the first operating mode, the first coupler clutch **164** is engaged. By engaging the first coupler clutch **164**, the first coupler assembly **158** and the second coupler assembly **160** operatively couple the third engine accessory rotor **118** to the second engine power transfer apparatus **122**. The second engine power transfer apparatus elements **138**, **140** and **142** may thereby rotate with the third engine accessory rotor **118**. Therefore, during this first operating mode, the third engine accessory rotor **118** may be rotationally driven by the second engine apparatus **146** (e.g., the low

speed rotating structure **60**) or the second engine accessory rotor **114**. Alternatively, the third engine accessory rotor **118** may drive rotation of (or mechanically boost power to) the second engine apparatus **146** (e.g., the low speed rotating structure **60**) and/or the second engine accessory rotor **114**.

(33) During the second operating mode, the first coupler clutch **164** is disengaged. The second engine power transfer apparatus elements **138**, **140** and **142** may rotate independent of the third engine accessory rotor **118**, and vice versa. Here, the third engine accessory **96** is operationally isolated from the operation of the engine core **46** (see FIG. **1**).

(34) In some embodiments, the mode of operation may be selected based on a rotational speed of the second engine apparatus **146**; e.g., the low speed rotating structure **60**. For example, when the second engine apparatus **146** is rotating at or below a threshold speed, the power coupler **90** may operate in the first operating mode. However, when the second engine apparatus **146** is rotating above the threshold speed, the power coupler **90** may operate in the second operating mode. With such operation, the power coupler **90** may operate in the second operating mode during higher power phases of aircraft flight such as aircraft takeoff, aircraft climb and aircraft cruise. By contrast, the power coupler **90** may operate in the first operating mode during lower power phases of aircraft flight such as aircraft descent, aircraft landing, aircraft taxiing and during startup of the aircraft powerplant **20** while on ground. The present disclosure, of course, is not limited to the foregoing exemplary power coupler schedule.

(35) In some embodiments, the power coupler **90** may be configured as a self-contained module. With such an arrangement, the power coupler **90** may have an internal lubrication system which is independent from a lubrication system **172** for the accessory gearbox **88**. Possible contaminants generated by wear of the first coupler clutch **164** may thereby be kept out of the accessory gearbox **88**, as well as other systems fluidly coupled to the accessory gearbox **88**. In addition, by independently mounting the power coupler **90** to the accessory gearbox **88**, the power coupler **90** may be removed from the accessory gearbox **88** as a single line replaceable unit (LRU) without requiring removal and/or disconnecting of aircraft powerplant components such as other engine accessories; e.g., **92**, **94** and/or **96**.

(36) In some embodiments, referring to FIG. **2**, the power coupler **90** and the third engine accessory **96** may be located to the opposing axial sides **106** and **104** of the accessory gearbox **88** and its gearbox housing **102**, respectively. With this arrangement, the power coupler **90** and the third engine accessory **96** may each be removably attached (e.g., mechanically fastened) to the exterior of the accessory gearbox **88** and its gearbox housing **102**. Moreover, the accessory gearbox **88** and its gearbox housing **102** (e.g., axially) physically separate the power coupler **90** from the third engine accessory **96**. In other embodiments, referring to FIGS. **3** and **4**, the power coupler **90** and the third engine accessory **96** may be located to a common one of the axial sides **104**, **106** of the accessory gearbox **88** and its gearbox housing **102**; e.g., the gearbox first side **104**. For example, the power coupler **90** of FIG. **3** is removably attached to the exterior of the accessory gearbox **88** at its gearbox first side **104**. The third engine accessory **96** is removably attached to an exterior of the power coupler **90** and its coupler housing **156**. With this arrangement, the third engine accessory **96** is indirectly mounted to the accessory gearbox **88** through the power coupler **90**. In another example, the power coupler **90** of FIG. **4** is integrated with the third engine accessory **96** into a module **174**; e.g., a single line-replaceable unit (LRU). More particularly, the power coupler members **158** and **160** and the third engine accessory rotor **118** of FIG. **4** are arranged within a common housing **176** (e.g., a case) for the power coupler-accessory module **174**.

(37) In some embodiments, referring to FIG. **2**, the power coupler **90** may be indirectly coupled to the third engine accessory **96** through the accessory power transfer apparatus **124**. In other embodiments, referring to FIGS. **3** and **4**, the power coupler **90** may be directly coupled to the third engine accessory **96**, and the accessory power transfer apparatus **124** of FIG. **2** may be omitted. The second coupler shaft **168** of FIGS. **3** and **4**, for example, is (e.g., directly) connected to and

rotatable with the third engine accessory rotor **118**. However, in other embodiments, it is contemplated one or more additional coupling elements may be provided between the second coupler shaft **168** and the third engine accessory rotor **118**. However, the additional coupling element(s) may be arranged outside of the accessory gearbox **88**; e.g., part of the power coupler **90**, the third engine accessory **96** and/or the power coupler-accessory module **174**.

(38) The power coupler **90** is generally described above with (a) the second engine power transfer apparatus **122** coupled to the power coupler **90** through the first coupler assembly **158** and (b) the third engine accessory rotor **118** coupled to the power coupler **90** through the second coupler assembly **160**. The present disclosure, however, is not limited to such an exemplary arrangement. For example, (a) the second engine power transfer apparatus **122** may alternatively be coupled to the power coupler **90** through the second coupler assembly **160** and (b) the third engine accessory rotor **118** may alternatively be coupled to the power coupler **90** through the first coupler assembly **158**.

(39) In some embodiments, referring to FIG. **5**, the third engine accessory **96** may be configured as an electric machine **178** that is electrically coupled to an electrical system **180** through an electric machine (EM) controller **182**. The electric machine **178** of FIG. **5** includes an electric machine rotor **184** (here, the third engine accessory rotor **118**), an electric machine stator **186** and an electric machine housing **188** (here, the third engine accessory housing **116**). The machine rotor **184** is rotatable about a rotational axis **190** of the machine rotor **184**, which rotational axis **190** may also be an axial centerline of the electric machine **178** and coaxial with a rotational axis of the second coupler assembly **160** and/or the accessory power transfer apparatus **124** (see FIG. **2**). The machine stator **186** of FIG. **5** is radially outboard of and circumscribes the machine rotor **184**. With this arrangement, each electric machine **178** is configured as a radial flux electric machine. The electric machine **178** of the present disclosure, however, is not limited to such an exemplary rotor-stator configuration nor to radial flux arrangements. The machine rotor **184**, for example, may alternatively be radially outboard of and circumscribe the machine stator **186**. In another example, the machine rotor **184** may be axially next to the machine stator **186** configuring the electric machine **178** as an axial flux electric machine. Referring again to FIG. **5**, the machine rotor **184** and the machine stator **186** are at least partially or completely housed within and interior of the machine housing **188**.

(40) Each electric machine **178** of FIG. **5** may be configurable as an electric motor and/or an electric generator; e.g., an electric motor-generator. For example, during a motor mode of operation, the electric machine **178** may operate as the electric motor to convert electricity received from the aircraft electrical system **180**. The machine stator **186**, for example, may generate an electromagnetic field with the machine rotor **184** using a current of electricity received from the aircraft electrical system **180** through the EM controller **182**. This electromagnetic field may drive rotation of the machine rotor **184**. The machine rotor **184**, in turn, may provide mechanical power to and drive rotation of the second engine apparatus **146** (see FIG. **2**) and/or other elements through the power coupler **90**. This mechanical power may be provided to boost power or completely power the rotation of the second engine apparatus **146** (see FIG. **2**) and/or other elements. By contrast, during a generator mode of operation, the electric machine **178** may operate as the electric generator to convert mechanical power received from the second engine apparatus **146** (see FIG. **2**) and/or other elements into electricity. Rotation of the machine rotor **184**, for example, may be rotationally driven by rotation of the second engine apparatus **146** (see FIG. **2**) and/or other elements through the power coupler **90**. The rotation of the machine rotor **184** may generate an electromagnetic field with the machine stator **186**, and the machine stator **186** may convert energy from the electromagnetic field into electricity. The electric machine **178** may then provide a current of electricity to the aircraft electrical system **180** through the EM controller **182** for storage and/or further use. The electric machine **178** of the present disclosure, however, is not limited to such exemplary operation. For example, the electric machine **178** may alternatively be configured as a

dedicated electric generator; e.g., without the electric motor functionality. In another example, the electric machine **178** may be configured as a dedicated electric motor; e.g., without the electric generator functionality.
(41) The EM controller **182** is electrically coupled to the electric machine **178** through one or more electric cables **192**; e.g., high voltage electric cables, power feeder cables, etc. More particularly, controller circuitry in the EM controller **182** is electrically coupled to the electric machine **178** and its machine stator **186** through the electric cables **192**. Similarly, the EM controller **182** is electrically coupled to an electrical distribution bus **194** of the aircraft electrical system **180** through one or more electric cables **195**; e.g., high voltage electric cables, power feeder cables, etc. More particularly, the controller circuitry in the EM controller **182** is electrically coupled to the aircraft electrical system **180** and its electrical distribution bus **194** through the respective electric cables **195**.
(42) The EM controller **182** and its controller circuitry are configured to control operation of the electric machine **178**. For example, when operating as the electric motor, the EM controller **182** is configured to regulate a flow of electricity from the aircraft electrical system **180** to the electric machine **178**. This electricity flow regulation may include: (a) turning-on the flow of electricity from the aircraft electrical system **180** to the electric machine **178** (e.g., electrically coupling the electric machine **178** to the aircraft electrical system **180**); (b) turning-off the flow of electricity from the aircraft electrical system **180** to the electric machine **178** (e.g., electrically decoupling the electric machine **178** from the aircraft electrical system **180**); (c) moderating the flow of electricity from the aircraft electrical system **180** to the electric machine **178**. Here, the EM controller **182** operates as a motor controller. In another example, when operating as the electric generator, the EM controller **182** is configured to regulate a flow of electricity from the electric machine **178** to the aircraft electrical system **180**. This electricity flow regulation may include: (a) turning-on the flow of electricity from the electric machine **178** to the aircraft electrical system **180** (e.g., electrically coupling the electric machine **178** to the aircraft electrical system **180**); (b) turning-off the flow of electricity from the electric machine **178** to the aircraft electrical system **180** (e.g., electrically decoupling the electric machine **178** from the aircraft electrical system **180**); (c) moderating the flow of electricity from the electric machine **178** to the aircraft electrical system **180**. Here, the EM controller **182** operates as a generator controller.
(43) The aircraft electrical system **180** includes the electrical distribution bus **194**. This aircraft electrical system **180** may also include a power source **196** and/or a power storage **198**. The electrical distribution bus **194** is electrically coupled to the electric machine **178** through the EM controller **182**. The electrical distribution bus **194** is also electrically coupled to the power source **196** and the power storage **198**. With this arrangement, the electrical distribution bus **194** provides an intermediate connection between the various electrical aircraft propulsion system members **182**, **196** and **198**. The power source **196** may be an electric generator powered by the turbine engine **24** (see FIG. **1**) or an electric generator powered by another aircraft powerplant; e.g., an engine of a companion aircraft propulsion system, an engine of an auxiliary power unit (APU), a fuel cell system, etc. The power storage **198** is configured to receive electricity from the electrical distribution bus **194** for storage. The power storage **198** is also configured to provide the stored electricity to the electrical distribution bus **194**. The power storage **198**, for example, may be configured as or otherwise include one or more electricity storage devices; e.g., batteries, super capacitors, etc.
(44) Referring to FIG. **2**, it is also contemplated one of the first engine accessories **92** (e.g., **92**A) may be configured as a dedicated electric machine (e.g., an electric motor-generator, an electric motor or an electric generator) associated with the first engine apparatus **134**. In addition or alternatively, it is contemplated the second engine accessory **94** may be configured as a dedicated electric machine (e.g., an electric motor-generator, an electric motor or an electric generator) associated with the second engine apparatus **146**.

(45) In some embodiments, referring to FIG. **1**, the accessory gearbox **88** may be arranged in the inner housing compartment **70**. The accessory gearbox **88**, for example, may be mounted to the inner case **66**. The present disclosure, however, is not limited to such an exemplary accessory gearbox location or mounting arrangement.
(46) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. An assembly for an aircraft powerplant, comprising: a gearbox including a first power transfer apparatus and a second power transfer apparatus independent of the first power transfer apparatus; and a power coupler mounted to an exterior of the gearbox and comprising a clutch; the first power transfer apparatus configured to operatively couple and transfer mechanical power through the gearbox between a first engine apparatus and an engine accessory; the power coupler configured to operatively couple a power coupled accessory to the second power transfer apparatus when the clutch is engaged, and the second power transfer apparatus and the power coupler collectively configured to operatively couple and transfer mechanical power through the gearbox between a second engine apparatus and the power coupled accessory when the clutch is engaged; and the power coupler configured to operatively decouple the second power transfer apparatus from the power coupled accessory when the clutch is disengaged.
2. The assembly of claim 1, wherein the power coupler is configured to engage the clutch when a rotational speed of the second engine apparatus is equal to or less than a threshold; and the power coupler is configured to disengage the clutch when the rotational speed of the second engine apparatus is greater than the threshold.
3. The assembly of claim 1, wherein the power coupler further comprises a gear system configured to transfer mechanical power between the power coupled accessory and the second power transfer apparatus when the clutch is engaged.
4. The assembly of claim 1, further comprising the power coupled accessory mounted to the gearbox through the power coupler.
5. The assembly of claim 1, further comprising the power coupled accessory mounted to the exterior of the gearbox independent of the power coupler.
6. The assembly of claim 5, wherein the power coupler is mounted to a first side of the gearbox; and the power coupled accessory is mounted to a second side of the gearbox opposite the first side of the gearbox.
7. The assembly of claim 1, wherein the gearbox further includes a third power transfer apparatus independent of the first power transfer apparatus and the second power transfer apparatus; and the third power transfer apparatus is configured to operatively couple and transfer mechanical power through the gearbox between the power coupler and the power coupled accessory.
8. The assembly of claim 7, wherein the third power transfer apparatus is configured as a pass-through shaft extending through the gearbox.
9. The assembly of claim 1, further comprising: the power coupled accessory; and a housing mounted to the exterior of the gearbox; the power coupler and the power coupled accessory housed within the housing.
10. The assembly of claim 1, further comprising: the power coupled accessory; the power coupler

and the power coupled accessory configured together in a single line replaceable unit that is removably mounted to the gearbox.

11. The assembly of claim 1, further comprising the power coupled accessory, wherein the power coupled accessory comprises an electric machine.

12. The assembly of claim 1, wherein the second power transfer apparatus is configured to operatively couple and transfer mechanical power through the gearbox between the second engine apparatus and a third engine accessory independent of the power coupler.

13. The assembly of claim 1, wherein the gearbox further includes a gearbox housing; and the second power transfer apparatus comprises a gear system disposed in the gearbox housing.

14. The assembly of claim 1, wherein the gearbox further includes a gearbox housing; the first power transfer apparatus comprises a first gear system disposed in the gearbox housing; and the second power transfer apparatus comprises a second gear system disposed in the gearbox housing.

15. The assembly of claim 1, further comprising: a compressor section, a combustor section, a first turbine section, a second turbine section and a flowpath extending through the compressor section, the combustor section, the first turbine section and the second turbine section from an inlet into the flowpath to an exhaust from the flowpath; the first engine apparatus comprising a first turbine rotor disposed in the first turbine section; and the second engine apparatus comprising a second turbine rotor disposed in the second turbine section.

16. An assembly for an aircraft powerplant, comprising: a first rotating structure comprising a first bladed engine rotor; a second rotating structure comprising a second bladed engine rotor; an engine accessory; a power coupled accessory; a gearbox including a first power transfer apparatus and a second power transfer apparatus that is mechanically disengaged from the first power transfer apparatus; and a power coupler between the power coupled accessory and the gearbox; the first power transfer apparatus configured to operatively couple and transfer mechanical power through the gearbox between the first rotating structure and the engine accessory; during a first operating mode, the power coupler configured to operatively couple the power coupled accessory to the second power transfer apparatus, and the second power transfer apparatus and the power coupler collectively configured to operatively couple and transfer mechanical power through the gearbox between the second rotating structure and the power coupled accessory; and during a second operating mode, the power coupler configured to operatively decouple the second power transfer apparatus from the power coupled accessory.

17. The assembly of claim 16, wherein the power coupler comprises a clutch configured to engage during the first operating mode; and disengage during the second operating mode.

18. The assembly of claim 16, wherein the power coupler is mounted to an exterior of the gearbox; and the power coupled accessory is mounted to an exterior of the power coupler.

19. The assembly of claim 16, wherein the power coupler is integrated with the power coupled accessory into a module that is removably coupled to an exterior of the gearbox.

20. An assembly for an aircraft powerplant, comprising: a first rotating structure comprising a first bladed engine rotor; a second rotating structure comprising a second bladed engine rotor; an engine accessory; a power coupled accessory; a gearbox including a first power transfer apparatus and a second power transfer apparatus that is mechanically disengaged from the first power transfer apparatus; and a power coupler arranged with the gearbox, wherein the gearbox is disposed between the power coupled accessory and the power coupler; the first power transfer apparatus configured to operatively couple and transfer mechanical power through the gearbox between the first rotating structure and the engine accessory; during a first operating mode, the power coupler configured to operatively couple the power coupled accessory to the second power transfer apparatus, and the second power transfer apparatus and the power coupler collectively configured to operatively couple and transfer mechanical power through the gearbox between the second rotating structure and the power coupled accessory; and during a second operating mode, the power

coupler configured to operatively decouple the second power transfer apparatus from the power coupled accessory.